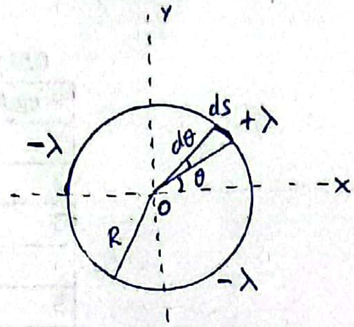
P1



a) $Q_{TOTAL} = Q(-\lambda) + Q(+\lambda) + Q(-\lambda)$

$$Q_{TOTAL} = \cancel{-\lambda\left(\frac{1}{4}2\pi R\right)} + \cancel{\lambda\left(\frac{1}{4}\times 2\pi R\right)}$$

$$-\lambda\left(\frac{1}{4}\times 2\pi R\right)$$

$$\rightarrow Q_{TOTAL} = -\frac{\lambda \pi R}{2}.$$

b) $V = k.\int \frac{dq}{r}$, $dq = \lambda\, ds = \lambda. R. d\theta$ y $r = R$

$$V_{+\lambda}(0) = \frac{k}{R}.\int_0^{\pi/2} \lambda R.\, d\theta = \frac{k\lambda.\pi}{2}$$

$$V_{-\lambda}(0) = -\frac{k}{R}.\int_{\pi/2}^{\pi} \lambda.R.\, d\theta = -\frac{k\lambda.\pi}{2}$$

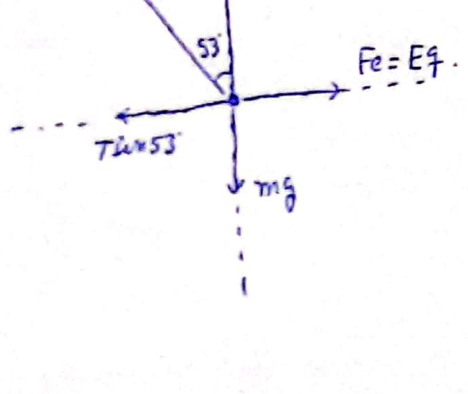
$$V_{-\lambda}(0) = -\frac{k}{R}\int_{-\pi/2}^{0} \lambda.R.\, d\theta = -\frac{k\lambda\pi}{2}$$

$$V_{TOTAL}(0) = -\frac{k\lambda\pi}{2}.$$

c) carga puntual $+q$ en O:

$$E_p = +q.V(0) = -\frac{k\lambda q\pi}{2} = -\frac{\lambda q}{8\epsilon_0}.$$

P2



a) $T.\left(\frac{3}{5}\right) = mg = 5\times10^{-6}(9,81)$

$T = 81,75\ \mu N.$

$\bar{T} = 81,75\,(-\,\text{sen}53\,\hat{i} + \cos 53\,\hat{j})\ \mu N$

$\vec{T} = (-\,65,4\,\hat{i} + 49,05\,\hat{j})\ \mu N.$

b) como los planos son infinitos y en el punto P el campo es nulo:

$$E_P = +\frac{\sigma_1}{2\epsilon_0} - \frac{|\sigma_2|}{2\epsilon_0} = 0 \rightarrow \sigma_1 = |\sigma_2| = \sigma$$

$\boxed{\text{En el punto T}}$: $E_T = -\frac{\sigma_1}{2\epsilon_0} + \frac{|\sigma_2|}{2\epsilon_0} \Rightarrow E_T = 0.$

$\boxed{\text{En el punto S}}$: $E_S = \frac{\sigma_1}{2\epsilon_0} + \frac{|\sigma_2|}{2\epsilon_0} = \frac{\sigma}{\epsilon_0}. \approx 4{,}09\ N/C$

c) Para la carga se tiene: $\left.\begin{array}{l} T\cos 53 = mg \\ T\sin 53 = Eq \end{array}\right\}$ $tg53° = \frac{Eq}{mg}$

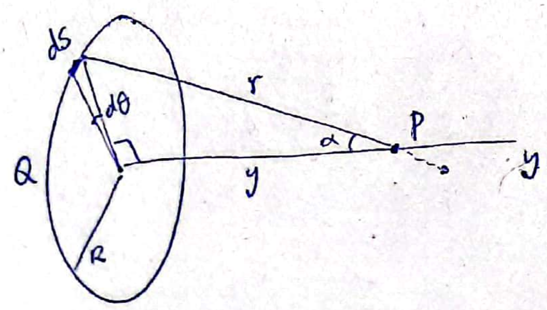
$$\rightarrow E = E_S = \left(\frac{mg}{q}\right) tg53° = \frac{\sigma}{\epsilon_0} \Rightarrow \sigma = \left(\frac{mg\,\epsilon_0}{q}\right) tg53°$$

$$\rightarrow \sigma = 3{,}62\times10^{-11}\ C/m^2 \quad \left\}\begin{array}{l} \sigma_1 = +\sigma \\ \sigma_2 = -\sigma \end{array}\right.$$

d) Para el potencial: $E = \frac{\Delta V}{d} \rightarrow \Delta V = Ed = \left(\frac{\sigma}{\epsilon_0}\right) d$

$$\underline{\Delta V = 0{,}41\ V.}$$

P3.



$$\lambda = \frac{Q}{2\pi R}$$

$$dq = \lambda R\, d\theta$$

$$V(y) = k\int\frac{dq}{r} = k.\int\frac{\lambda R\, d\theta}{r} = \frac{k\lambda R}{r}\int_0^{2\pi} d\theta = \frac{k\lambda R}{r}(2\pi) = \frac{2\pi k\lambda R}{\sqrt{R^2+y^2}}$$

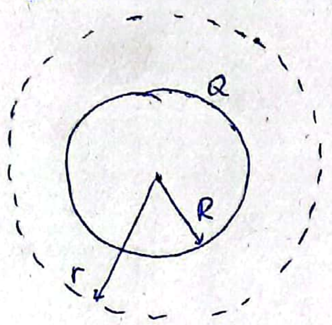
$$V(y) = \frac{kQ}{\sqrt{R^2+y^2}} \rightarrow V(y=1) = \frac{9\times10^9 \times 12\times10^{-6}\times 2\pi\times 1{,}5}{\sqrt{1^2+1{,}5^2}} = \underline{5{,}65\times10^5\ V}$$

$$E(y) = k \cdot \int \frac{dq}{r^2} \cdot \cos\alpha = k \int_0^{2\pi} \frac{\lambda R \, d\theta}{r^2} \cdot \cos\alpha = \frac{k\lambda R}{r^2} \cdot (2\pi) \cdot \cos\alpha = \frac{2\pi k \lambda R}{R^2+y^2} \cdot \cos\alpha$$

$$E(y) = \frac{2\pi \cdot K \cdot \lambda R \cdot y}{\sqrt{R^2+y^2}^{\,3}} = \frac{k \cdot y \cdot Q}{(R^2+y^2)^{3/2}}$$

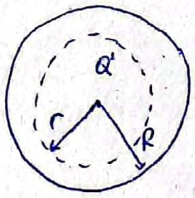
$$E(y=1m) = \frac{9\times 10^{9} \times 1 \times 12\times 10^{-6} \times 2\pi \times 1{,}5}{(1^2+1{,}5^2)^{3/2}} = 17{,}37\times 10^{4}\ N/C$$

P4. Cálculo del campo eléctrico:



$$E_{ext} = \frac{Q}{4\pi\epsilon_0 r^2}, \quad Q = \int \rho \, dv = 4\pi \int_0^R (\alpha r) r^2 dr = \pi\alpha R^4$$

$$\rightarrow E_{ext} = \frac{\alpha R^4}{4\epsilon_0 r^2}, \quad r > R.$$



$$E_{int} = \frac{Q'}{4\pi\epsilon_0 r^2}, \quad Q' = \int \rho . dv' = 4\pi \int_0^r (\alpha r) r^2 dr = \pi\alpha r^4$$

$$\rightarrow E_{int} = \frac{\alpha r^2}{4\epsilon_0}, \quad r < R.$$

Energía eléctrica: $U = \frac{\epsilon_0}{2} \int_{\text{Todo el espacio}} E^2 . dv$

$$U = \frac{\epsilon_0}{2}\int_0^R E_{int}^2 \, dv + \frac{\epsilon_0}{2}\int_R^\infty E_{ext}^2 \, dv = 2\pi\epsilon_0 \left\{ \int_0^R \left(\frac{\alpha^2 r^4}{16\epsilon_0^2}\right) r^2 dr + \int_R^\infty \left(\frac{\alpha^2 R^8}{16\epsilon_0^2 r^4}\right) r^2 dr \right\}$$

$$U = \frac{\pi\alpha^2 . R^7}{7\epsilon_0}.$$